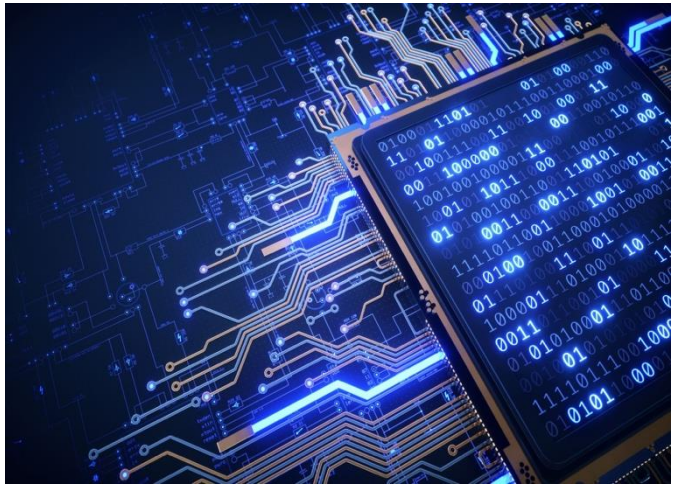
AI Components

Knowledge Representation



Reasoning

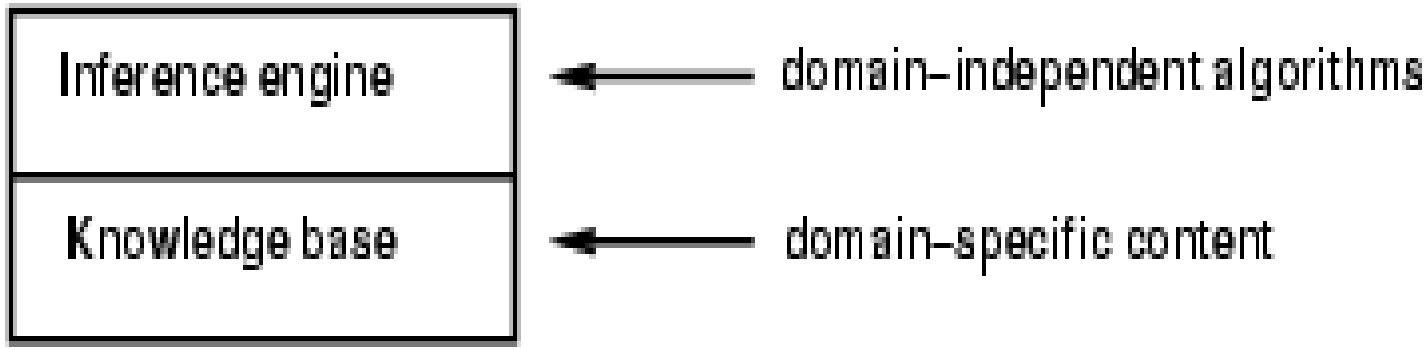
Search & Planning

Decision Making

Machine Learning

Knowledge Base Agents

- Knowledge base = set of sentences in a formal language
- Declarative approach to building an agent (or other system):
  - `Tell` it what it needs to know
- Then it can `Ask` itself what to do - answers should follow from the KB

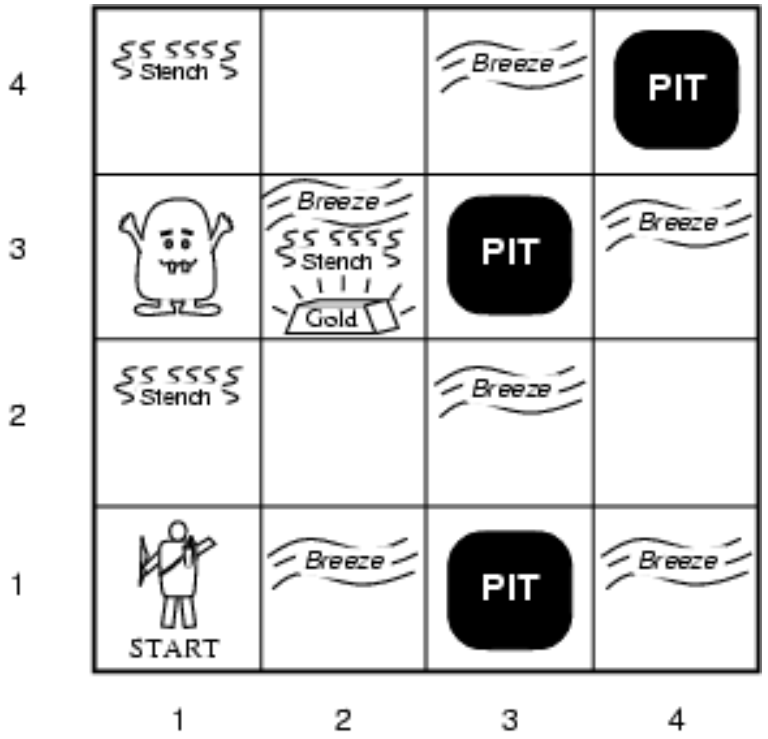


Knowledge Base Agents

- The agent must be able to:
  - Represent states, actions, etc.
  - Incorporate new percepts
  - Update internal representations of the world
  - Deduce hidden properties of the world
  - Deduce appropriate actions

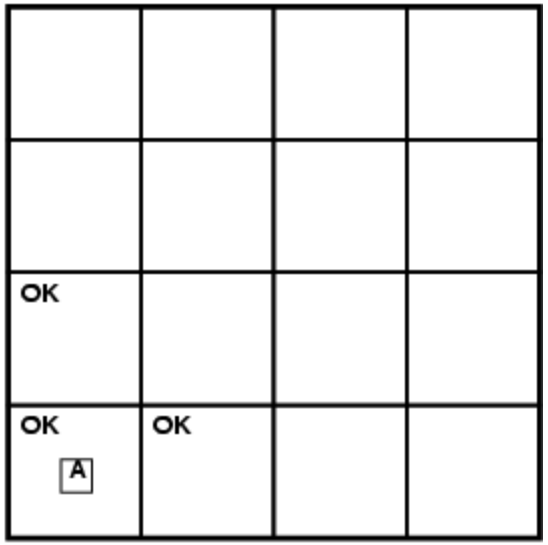
Example: Wumpus World

- Performance measure
  - gold +1000, death -1000
  - -1 per step, -10 for using the arrow
- Environment
  - Squares adjacent to wumpus are smelly
  - Squares adjacent to pit are breezy
  - Glitter iff gold is in the same square
  - Shooting kills wumpus if you are facing it
  - Shooting uses up the only arrow
  - Grabbing picks up gold if in same square
  - Releasing drops the gold in same square

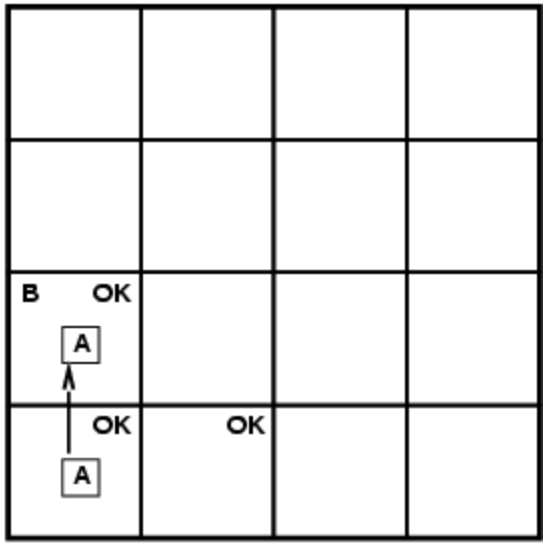


- Sensors: Stench, Breeze, Glitter, Bump (when walk into a wall), Scream (when wumpus is killed)
- Actuators: Left turn, Right turn, Forward, Grab, Release, Shoot

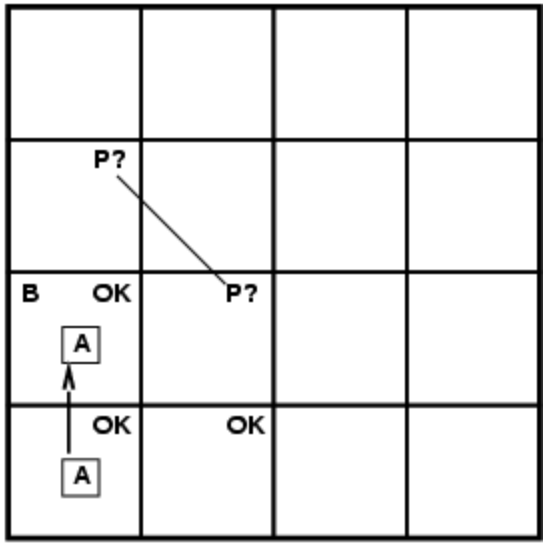
Exploring a wumpus world



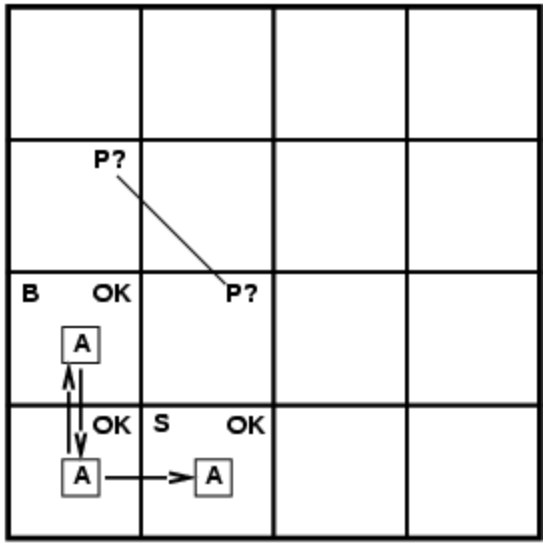
Exploring a wumpus world



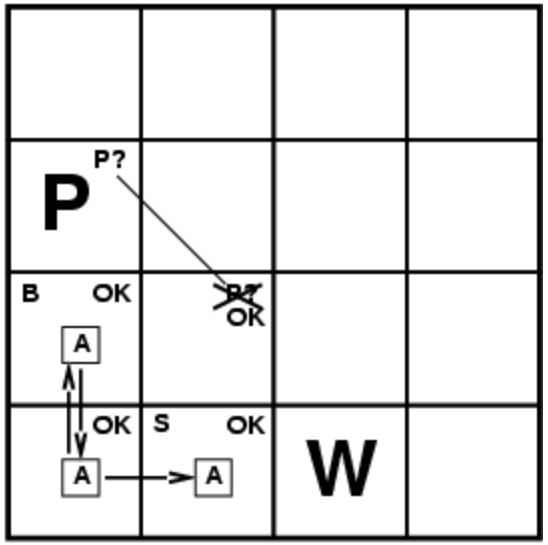
Exploring a wumpus world



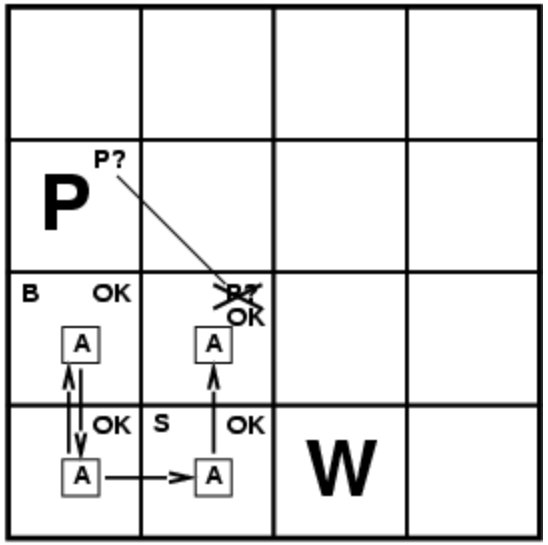
Exploring a wumpus world



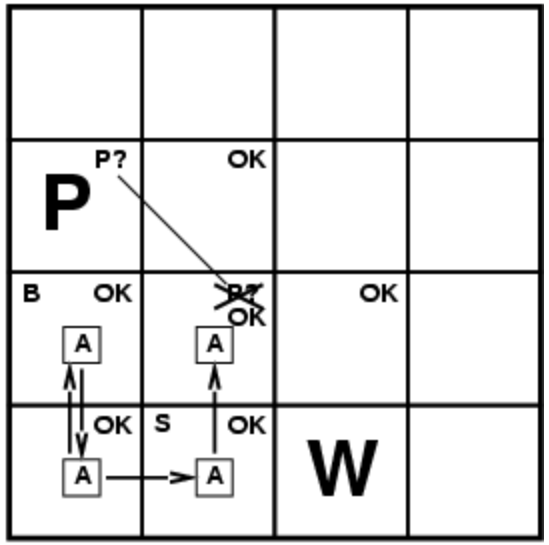
Exploring a wumpus world



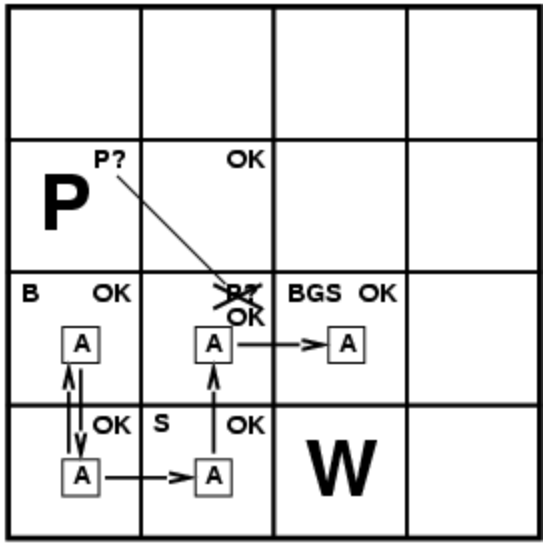
Exploring a wumpus world



Exploring a wumpus world



Exploring a wumpus world



Logic in general

- Logics are formal languages for representing information such that conclusions can be drawn
- Syntax defines the sentences in the language
- Semantics define the "meaning" of sentences;
  - i.e., define truth of a sentence in a world

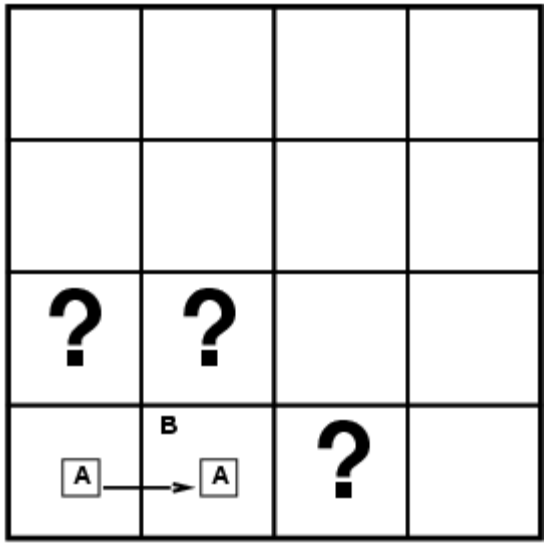
- E.g., the language of arithmetic
  - $x+2 \geq y$ is a sentence; $x2+y > \{\}$ is not a sentence
  - $x+2 \geq y$ is true iff the number $x+2$ is no less than the number y
  - $x+2 \geq y$ is true in a world where $x = 7, y = 1$
  - $x+2 \geq y$ is false in a world where $x = 0, y = 6$

Entailment in the wumpus world

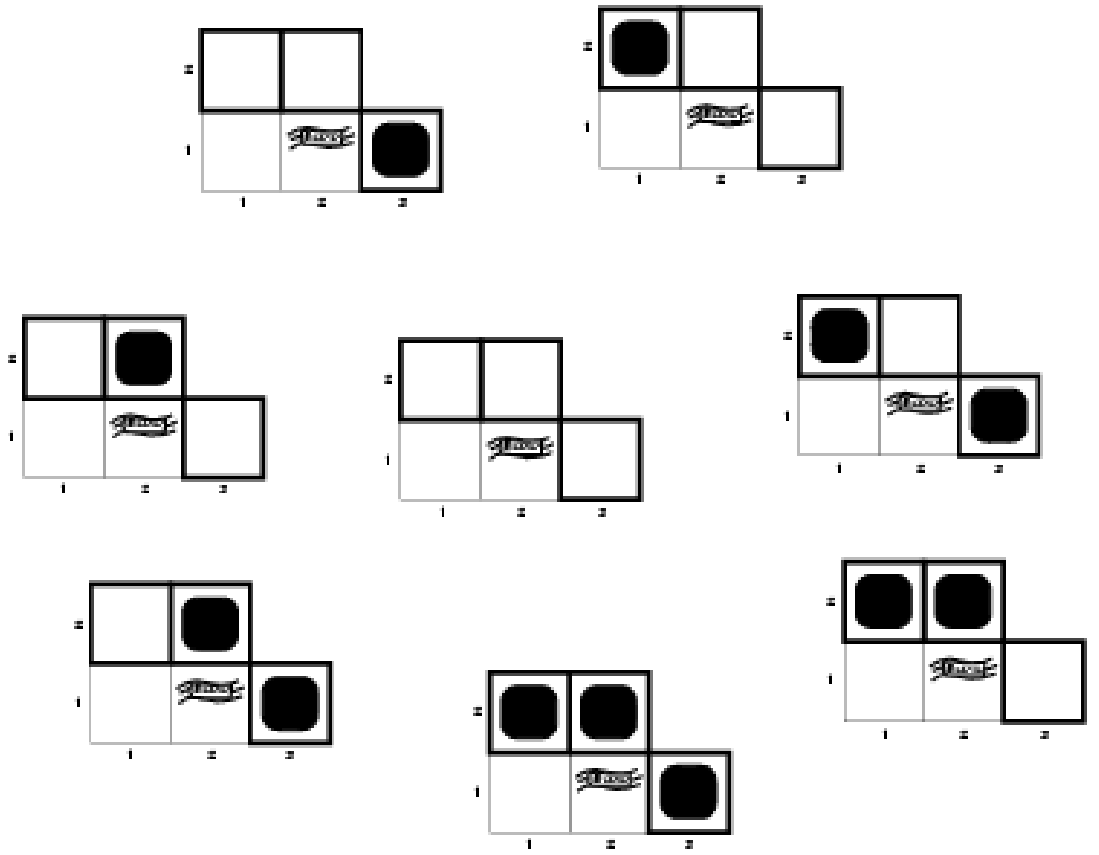
Situation after detecting nothing in [1,1], moving right, breeze in [2,1]

Consider possible models for *KB* assuming only pits

3 Boolean choices $\Rightarrow$ 8 possible models



Wumpus models



Propositional logic: Syntax

- Propositional logic is the simplest logic – illustrates basic ideas

- The proposition symbols P_1, P_2 etc are sentences

  - If S is a sentence, $\neg$S is a sentence (negation)
  - If S_1 and S_2 are sentences, $S_1 \wedge S_2$ is a sentence (conjunction)
  - If S_1 and S_2 are sentences, $S_1 \vee S_2$ is a sentence (disjunction)
  - If S_1 and S_2 are sentences, $S_1 \Rightarrow S_2$ is a sentence (implication)
  - If S_1 and S_2 are sentences, $S_1 \Leftrightarrow S_2$ is a sentence (biconditional)

Propositional logic: Semantics

- Each model specifies true/false for each proposition symbol

  E.g. $P_{1,2}$ $P_{2,2}$ $P_{3,1}$
  false true false

- With these symbols, 8 possible models, can be enumerated automatically.
- Rules for evaluating truth with respect to a model *m*:

$\neg S$	is true iff	S is false	
$S_1 \wedge S_2$	is true iff	S_1 is true and	S_2 is true
$S_1 \vee S_2$	is true iff	$S_1$is true or	S_2 is true
$S_1 \Rightarrow S_2$	is true iff	S_1 is false or	S_2 is true
$S_1 \Leftrightarrow S_2$	is true iff	$S_1 \Rightarrow S_2$ is true and	$S_2 \Rightarrow S_1$ is true

- Simple recursive process evaluates an arbitrary sentence, e.g.,

$$\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1}) = \mathit{true} \wedge (\mathit{true} \vee \mathit{false}) = \mathit{true} \wedge \mathit{true} = \mathit{true}$$

Truth tables for connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
$false$	$false$	$true$	$false$	$false$	$true$	$true$
$false$	$true$	$true$	$false$	$true$	$true$	$false$
$true$	$false$	$false$	$false$	$true$	$false$	$false$
$true$	$true$	$false$	$true$	$true$	$true$	$true$

Wumpus world sentences

Let $P_{i,j}$ be true if there is a pit in [i, j].

Let $B_{i,j}$ be true if there is a breeze in [i, j].

$\neg P_{1,1}$

$\neg B_{1,1}$

$B_{2,1}$

- "Pits cause breezes in adjacent squares"

$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

$B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

Truth tables for inference

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	KB	α_1
$false$	$false$	$false$	$false$	$false$	$false$	$false$	$false$	$true$
$false$	$false$	$false$	$false$	$false$	$false$	$true$	$false$	$true$
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$false$	$true$	$false$	$false$	$false$	$false$	$false$	$false$	$true$
$false$	$true$	$false$	$false$	$false$	$false$	$true$	$true$	$true$
$false$	$true$	$false$	$false$	$false$	$true$	$false$	$true$	$true$
$false$	$true$	$false$	$false$	$false$	$true$	$true$	$true$	$true$
$false$	$true$	$false$	$false$	$true$	$false$	$false$	$false$	$true$
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$true$	$true$	$true$	$true$	$true$	$true$	$true$	$false$	$false$

Inference : Forward-chaining and Backward-chaining

ÖZYEĞİN ÜNİVERSİTESİ

- Horn Form (restricted)

  KB = conjunction of Horn clauses

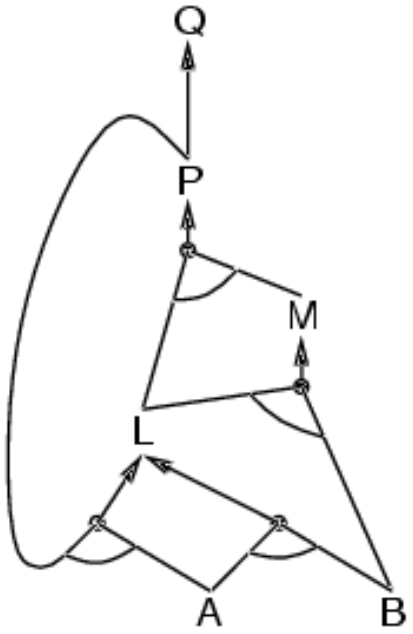
  - Horn clause = Disjunction of literals of which at most one is positive
    - E.g. $(\neg B \vee \neg Q \vee R)$ is horn clause but $(\neg B \vee Q \vee R)$ is not a horn clause.
  - Horn clause can be written as implications:
    - (conjunction of symbols) $\Rightarrow$ symbol
    - E.g. $B \wedge Q \Rightarrow R$ (premise called body and the conclusion is called head).

- Inference with Horn clauses can be done through forward-chaining and backward-chaining algorithms.

Inference: Forward-chaining

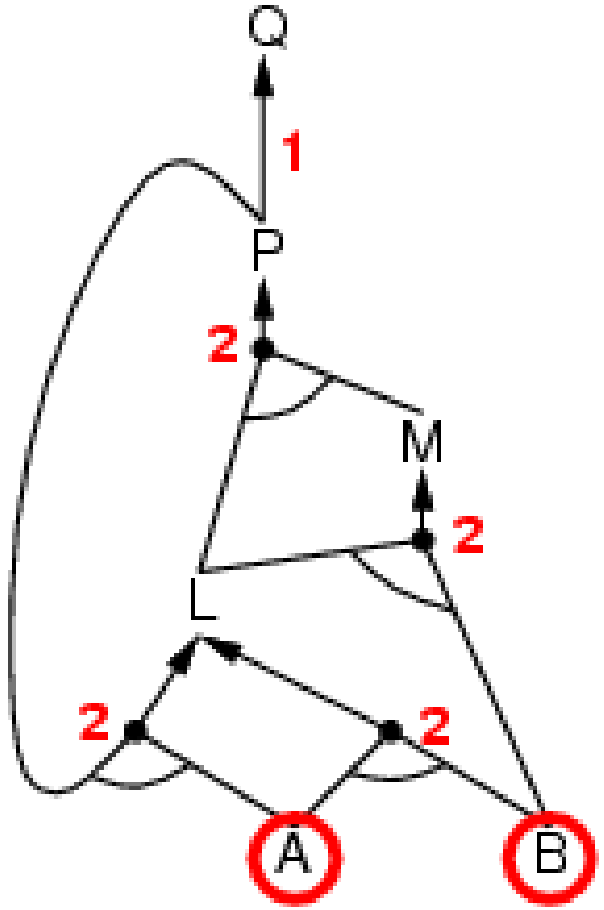
- Idea:
  - fire any rule whose premises are satisfied in the *KB*,
  - add its conclusion to the *KB*, until query is found

$$P \Rightarrow Q$$
$$L \wedge M \Rightarrow P$$
$$B \wedge L \Rightarrow M$$
$$A \wedge P \Rightarrow L$$
$$A \wedge B \Rightarrow L$$
$$A$$
$$B$$



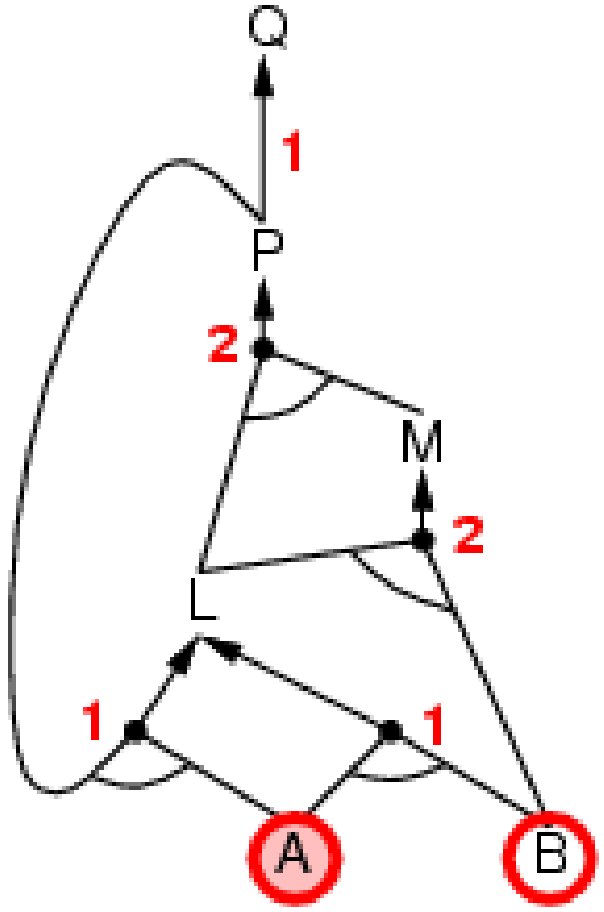
Forward chaining example

$P \Rightarrow Q$
$L \wedge M \Rightarrow P$
$B \wedge L \Rightarrow M$
$A \wedge P \Rightarrow L$
$A \wedge B \Rightarrow L$
A
B



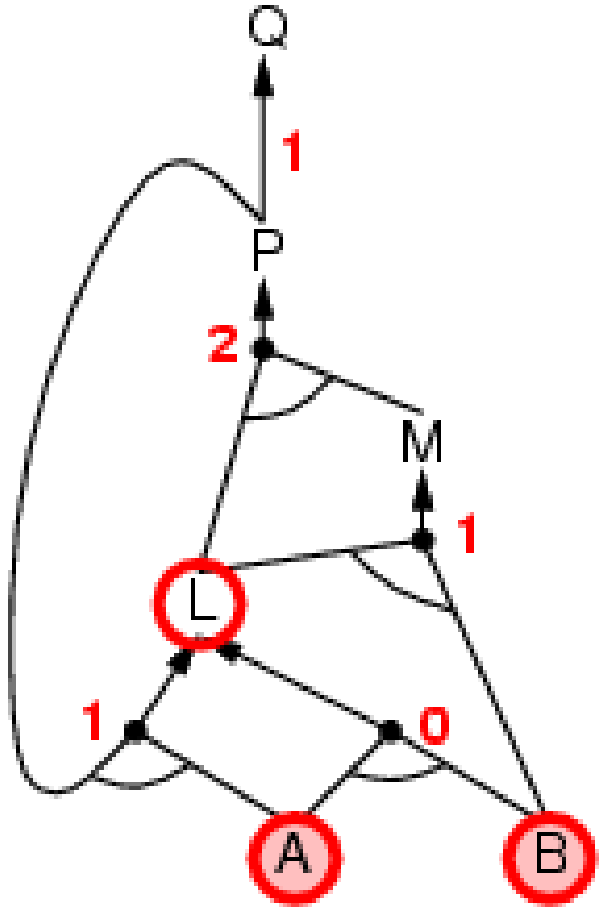
Forward chaining example

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$L \wedge M \Rightarrow P$
$B \wedge L \Rightarrow M$
$A \wedge P \Rightarrow L$
$A \wedge B \Rightarrow L$
A
B



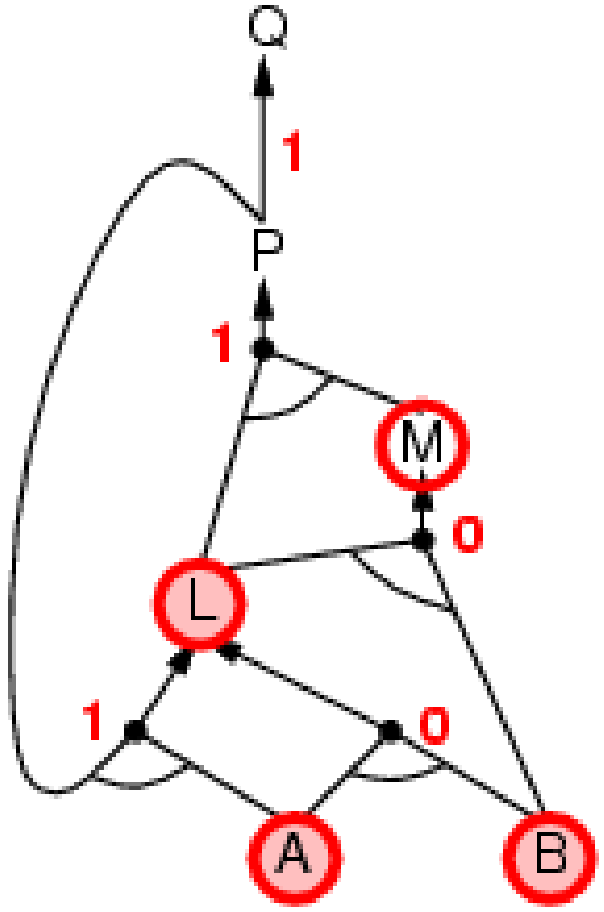
Forward chaining example

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$L \wedge M \Rightarrow P$
$B \wedge L \Rightarrow M$
$A \wedge P \Rightarrow L$
$A \wedge B \Rightarrow L$
A
B



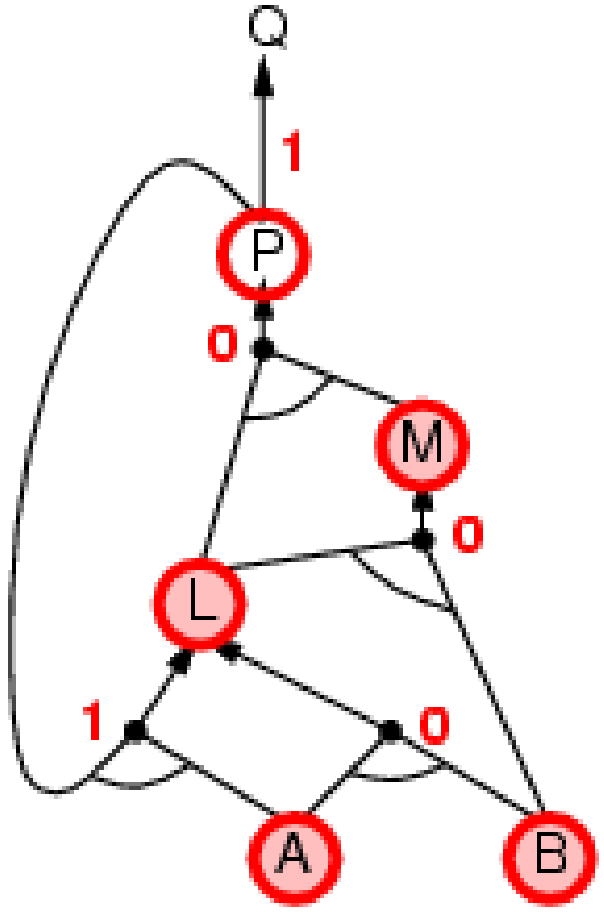
Forward chaining example

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$L \wedge M \Rightarrow P$
$B \wedge L \Rightarrow M$
$A \wedge P \Rightarrow L$
$A \wedge B \Rightarrow L$
A
B



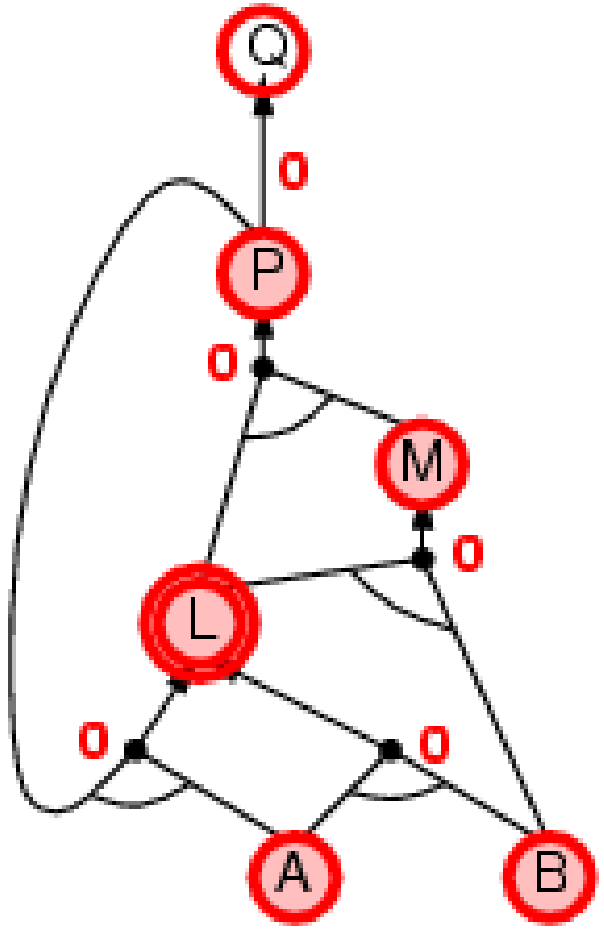
Forward chaining example

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$L \wedge M \Rightarrow P$
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A
B



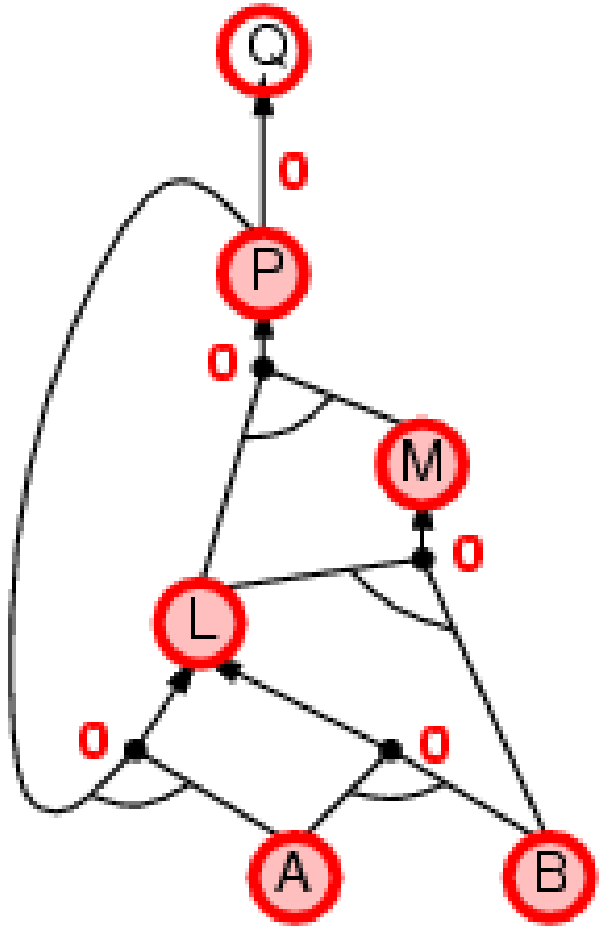
Forward chaining example

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$A \wedge P \Rightarrow L$
$A \wedge B \Rightarrow L$
A
B



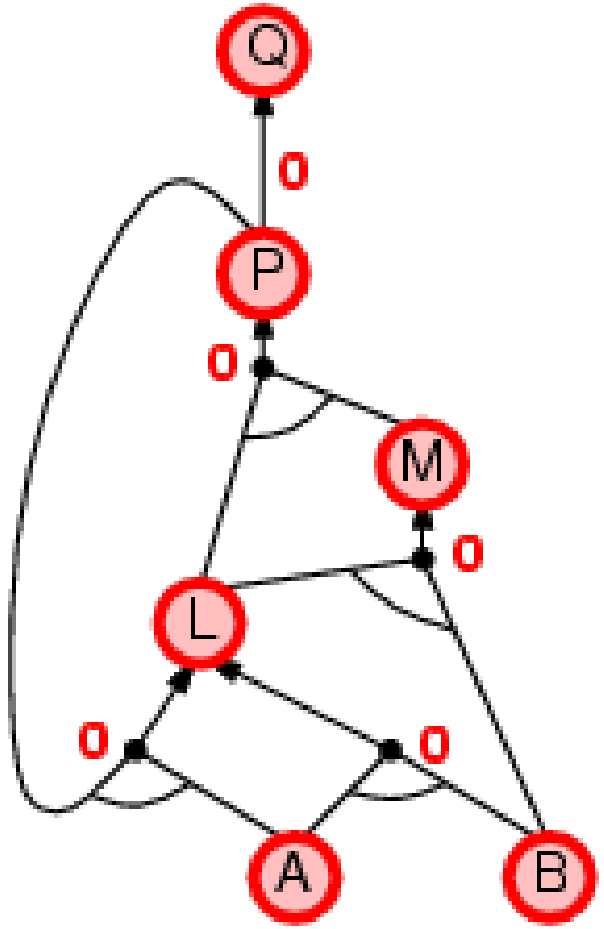
Forward chaining example

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A
B



Forward chaining example

$P \Rightarrow Q$
$L \wedge M \Rightarrow P$
$B \wedge L \Rightarrow M$
$A \wedge P \Rightarrow L$
$A \wedge B \Rightarrow L$
A
B



Inference: Backward chaining

Idea: work backwards from the query *q*:

To prove *q* by Backward chaining,

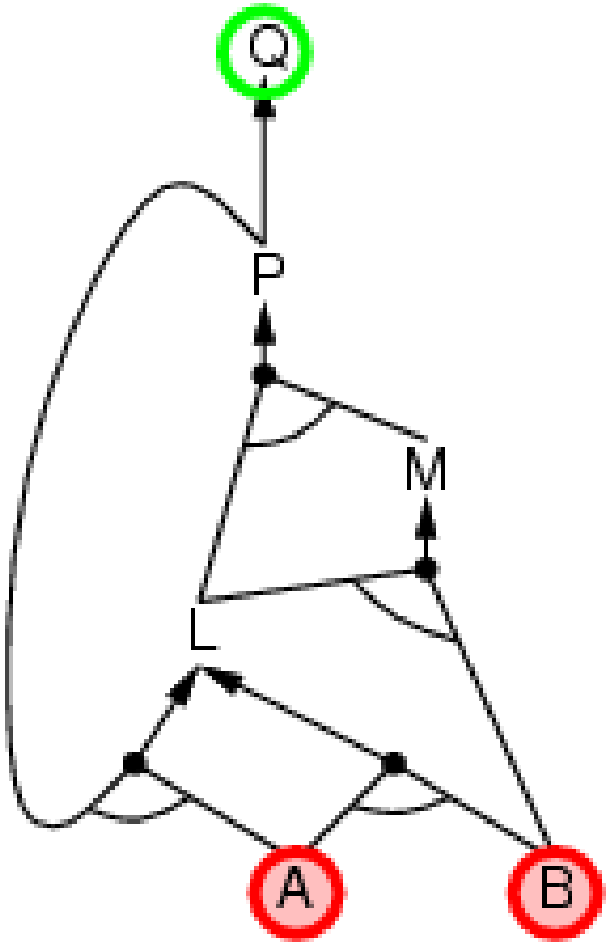
(1) check if *q* is known already → no work is needed

(2) finds implications whose conclusion is q.

(3) If all premises of one of those implications can be true (backward chaining), then q is true.

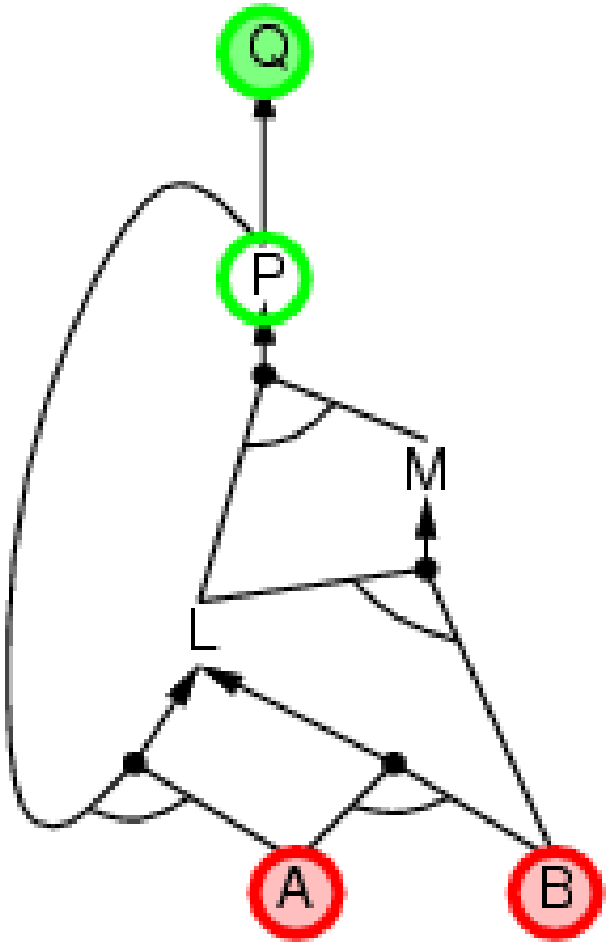
Backward chaining example

$P \Rightarrow Q$
$L \wedge M \Rightarrow P$
$B \wedge L \Rightarrow M$
$A \wedge P \Rightarrow L$
$A \wedge B \Rightarrow L$
A
B



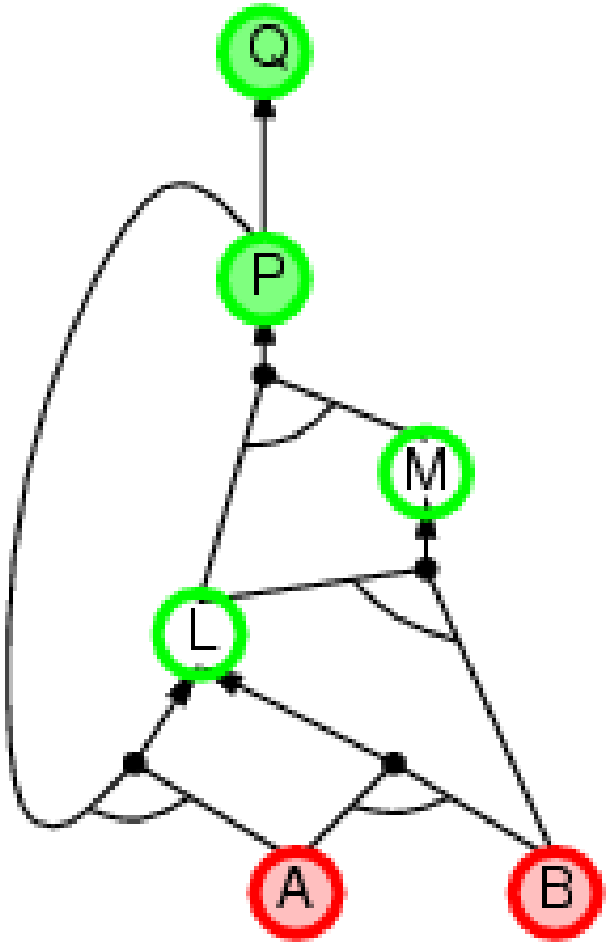
Backward chaining example

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A
B



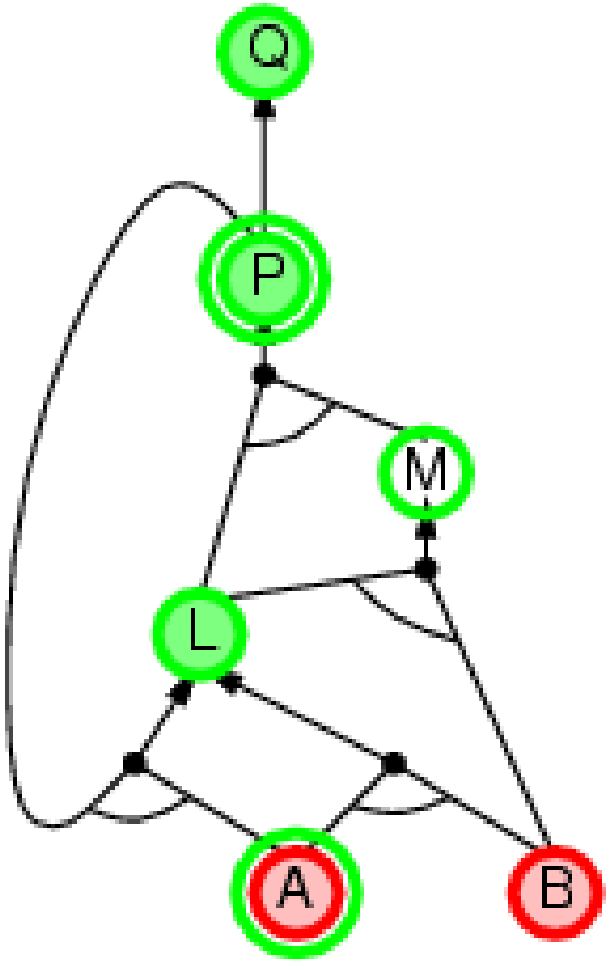
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A
B



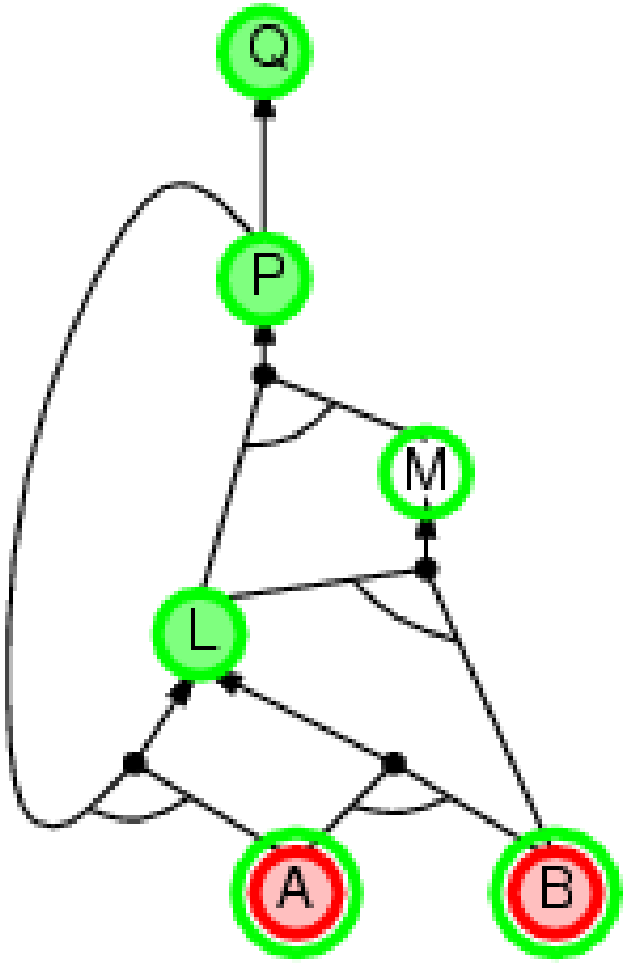
Backward chaining example

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A
B



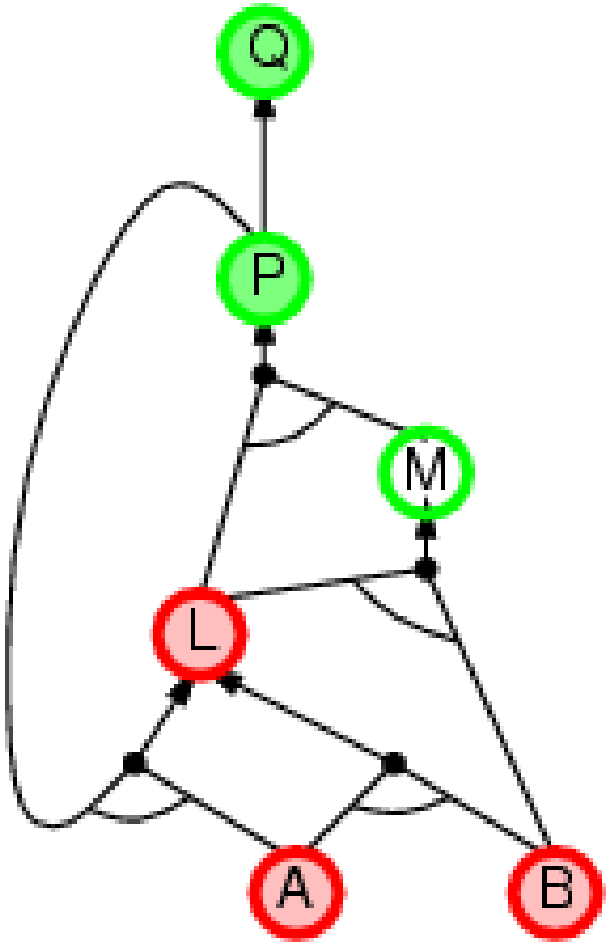
Backward chaining example

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A
B



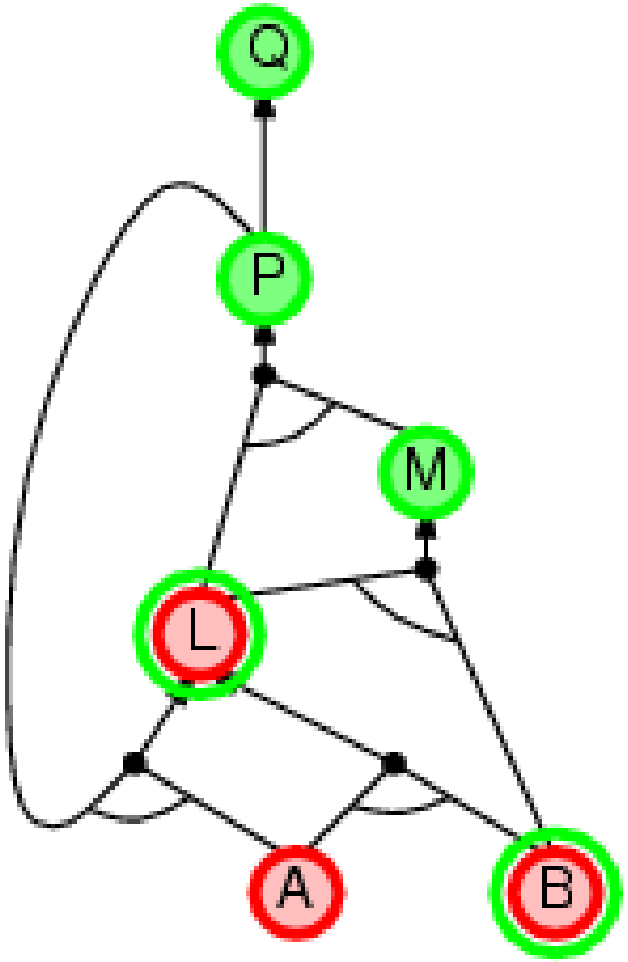
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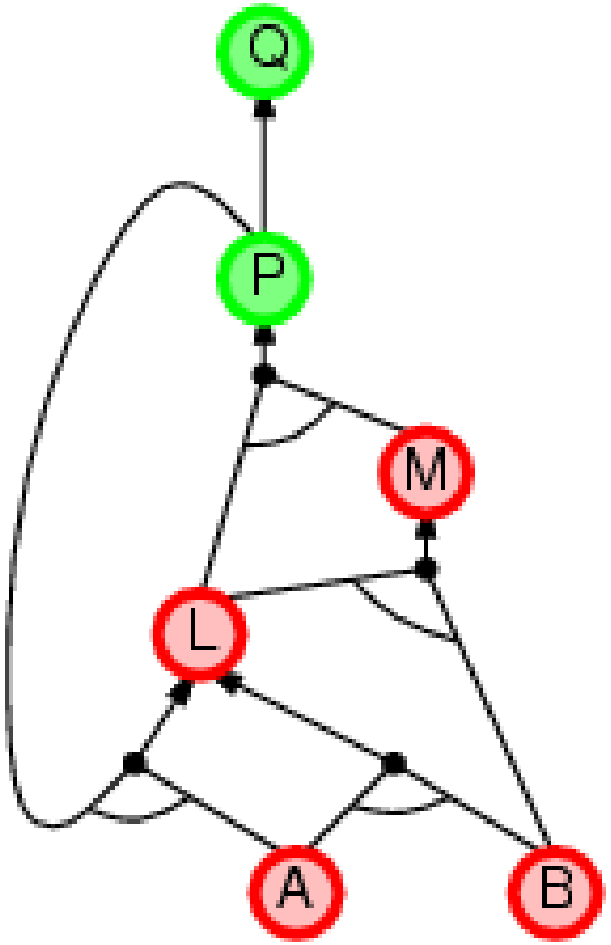
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B



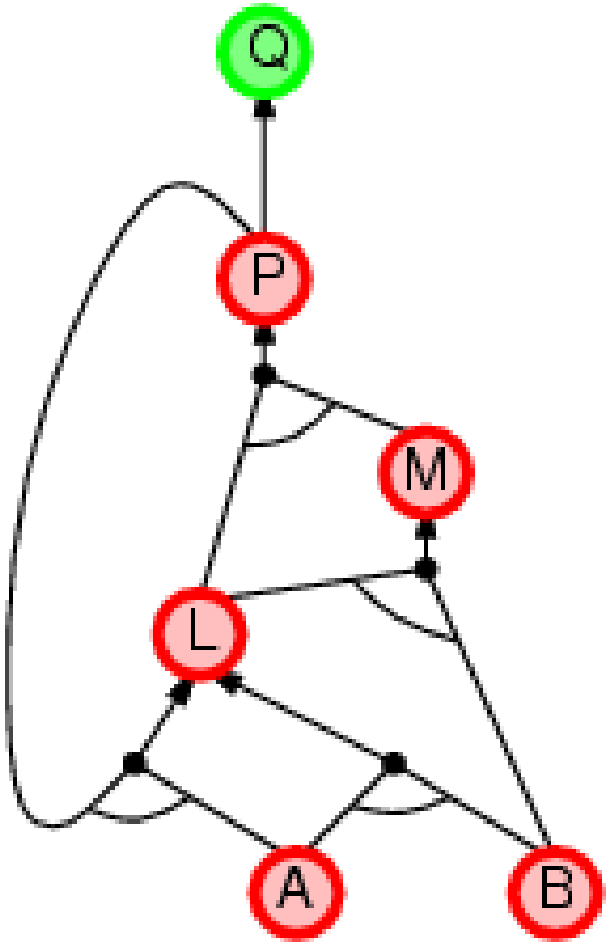
Backward chaining example

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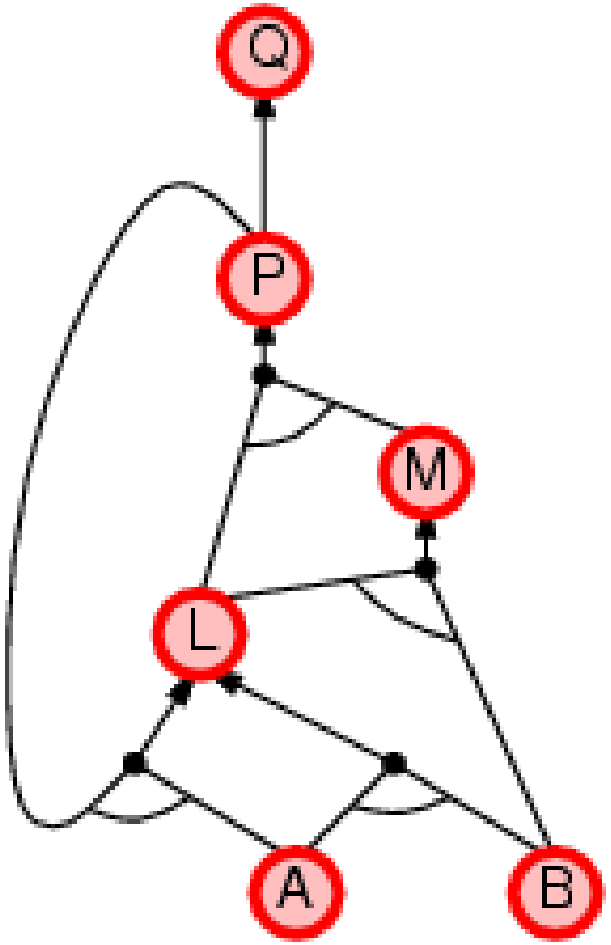
Backward chaining example

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B



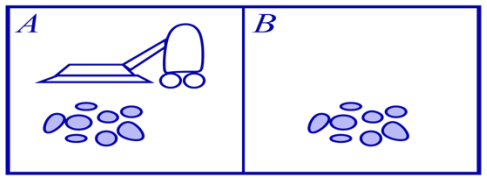
Backward chaining example

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$A \wedge P \Rightarrow L$
$A \wedge B \Rightarrow L$
A
B



Propositional Logic

- Propositional Logic is a powerful language but it has some limitations:
  - No capability to handle uncertainty
  - We can talk about event that are true or false in the world but not about objects having some properties such as color, size, weight etc. Nor can we talk about relations about objects.
  - There are no shortcuts to succinctly talk about a lot of different things happening.
  - For example, if we have a vacuum-cleaner world with thousands of location, and we want to say that every location is free of dirt. We need a conjunction of 1000 propositions.



Introduction to AI course in https://classroom.udacity.com/courses/cs271

First-Order Logic

Some Examples of FOL using quantifier:

1. All birds fly.

In this question the predicate is "**fly(bird)**."

And since there are all birds who fly so it will be represented as follows.

$\forall x$ **bird(x)** $\rightarrow$**fly(x)**.

2. Every man respects his parent.

In this question, the predicate is "**respect(x, y)," where x=man, and y= parent**.

Since there is every man so will use $\forall$, and it will be represented as follows:

$\forall x$ **man(x)** $\rightarrow$ **respects (x, parent)**.

3. Some boys play cricket.

In this question, the predicate is "**play(x, y),**" where x= boys, and y= game. Since there are some boys so we will use $\exists$**, and it will be represented as**:

$\exists x$ **boys(x)** $\rightarrow$ **play(x, cricket)**.

4. Not all students like both Mathematics and Science.

In this question, the predicate is "**like(x, y)," where x= student, and y= subject**.

Since there are not all students, so we will use $\forall$ **with negation, so** following representation for this:

$\neg\forall$ **(x) [student(x)** $\rightarrow$ **like(x, Mathematics)** $\wedge$ **like(x, Science)].**

5. Only one student failed in Mathematics.

In this question, the predicate is "**failed(x, y)," where x= student, and y= subject**.

Since there is only one student who failed in Mathematics, so we will use following representation for this:

$\exists$**(x) [student(x)** $\rightarrow$ **failed (x, Mathematics)** $\wedge\forall$ **(y) [**$\neg$**(x==y)** $\wedge$ **student(y)** $\rightarrow$ $\neg$**failed (x, Mathematics)]**.

https://www.javatpoint.com/first-order-logic-in-artificial-intelligence

Example knowledge base

- The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American.

Example knowledge base contd.

... it is a crime for an American to sell weapons to hostile nations:

$American(x) \wedge Weapon(y) \wedge Sells(x,y,z) \wedge Hostile(z) \Rightarrow Criminal(x)$

Nono … has some missiles, i.e.,

$\exists x\ Owns(Nono,x) \wedge Missile(x)$:

$Owns(Nono,M_1)$ and $Missile(M_1)$

… all of its missiles were sold to it by Colonel West

$Missile(x) \wedge Owns(Nono,x) \Rightarrow Sells(West,x,Nono)$

Missiles are weapons:

$Missile(x) \Rightarrow Weapon(x)$

An enemy of America counts as "hostile“:

$Enemy(x,America) \Rightarrow Hostile(x)$

West, who is American …

$American(West)$

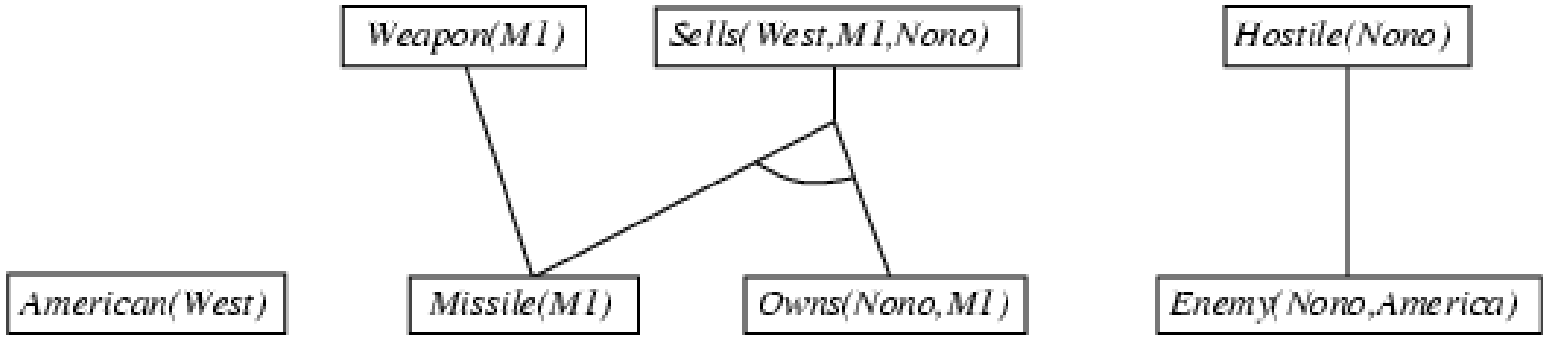
The country Nono, an enemy of America …

$Enemy(Nono,America)$

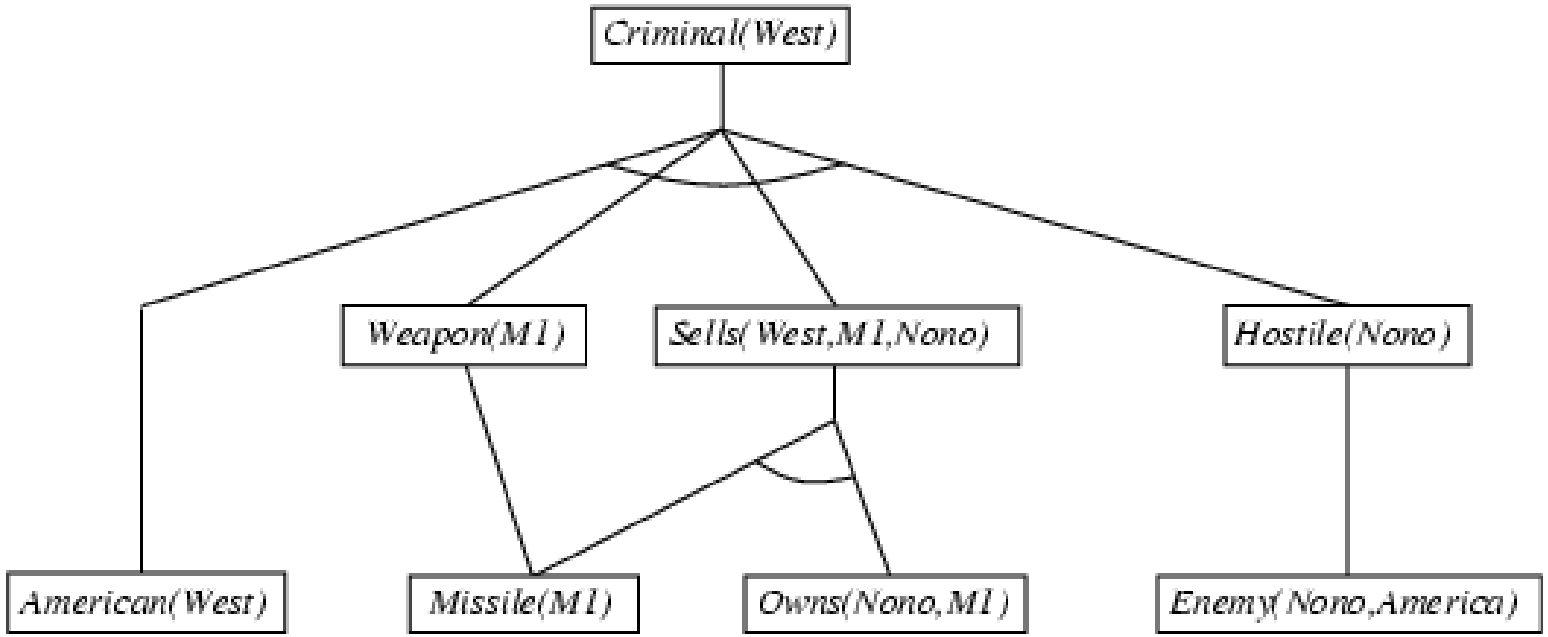
Forward chaining proof



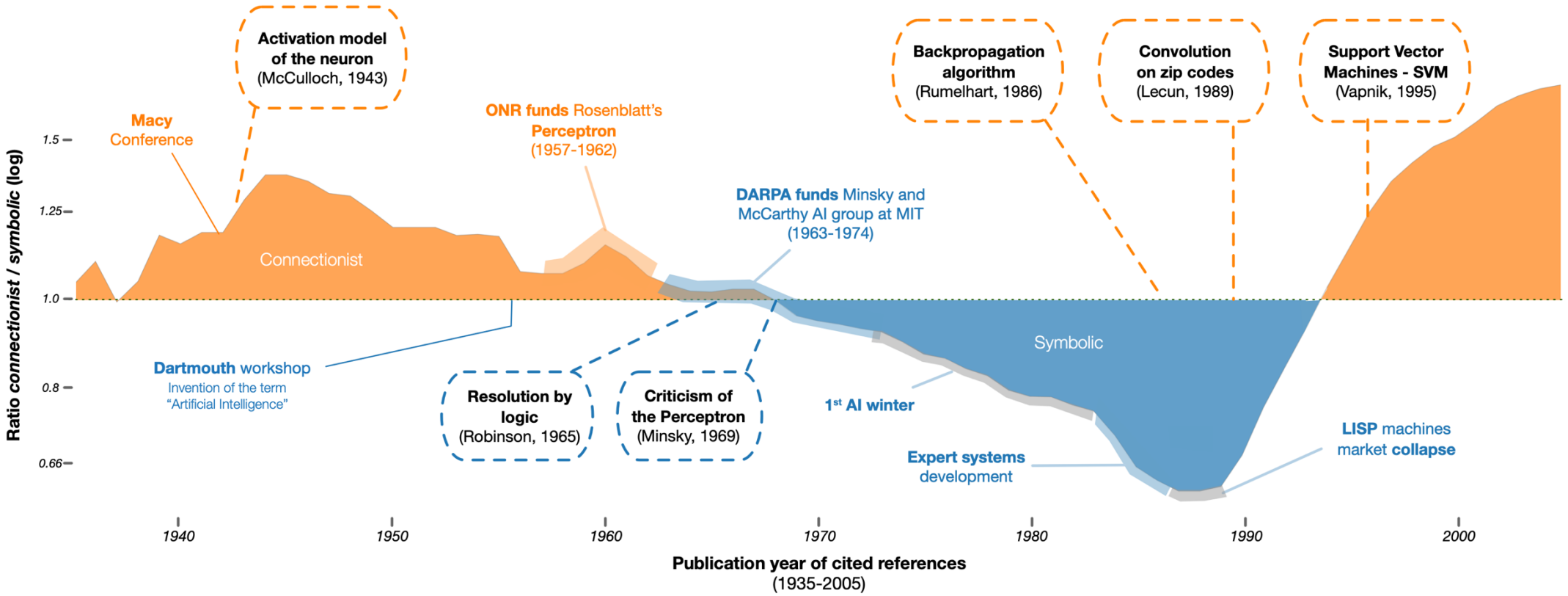
Forward chaining proof

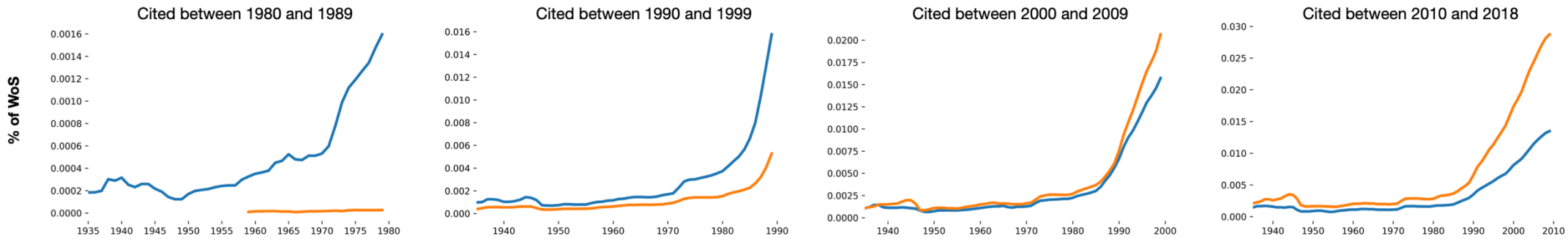


Forward chaining proof



Symbolic vs Connectionist





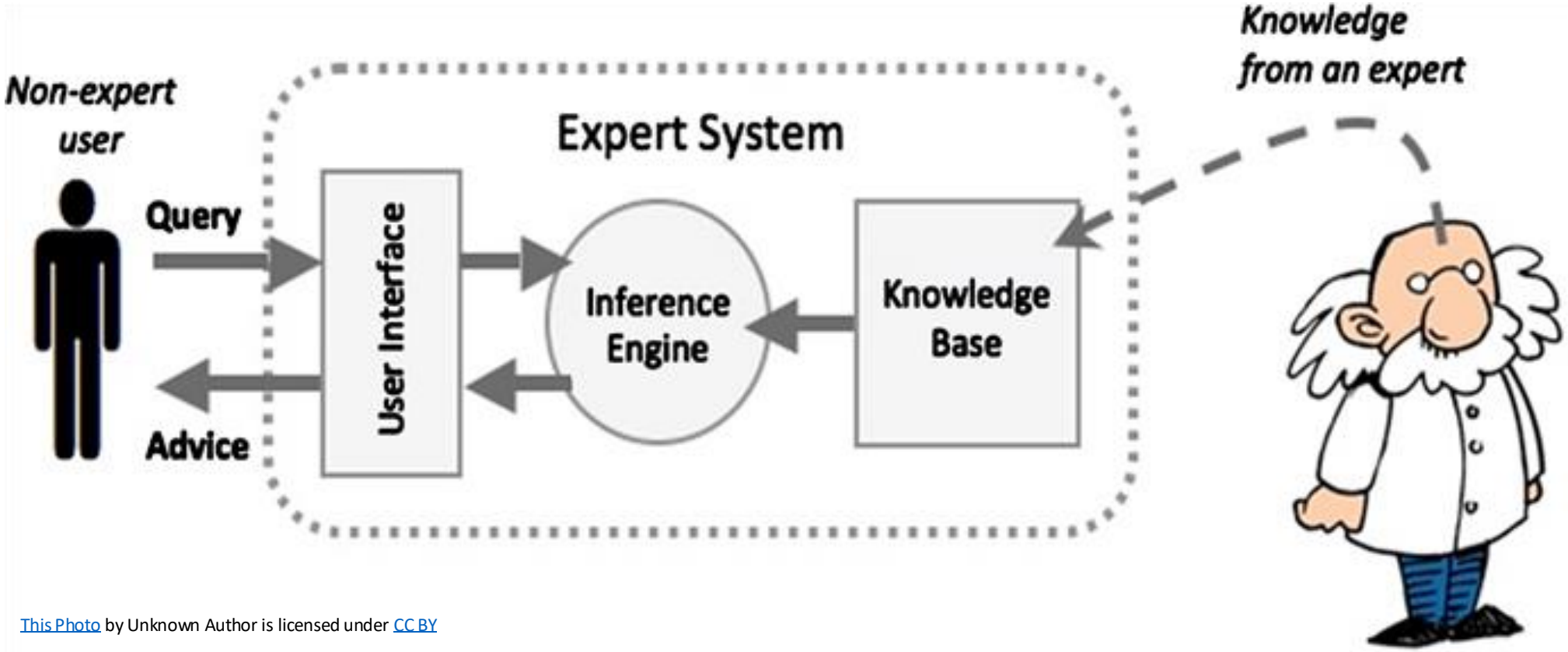
John Abela, LLM Lecture

What is an Expert System?

- Think of an expert system as a way of storing the knowledge of a doctor, lawyer, or financial advisor
- Whenever you need, you can interact with such a system !

"An expert system is a sophisticated computer program designed to **mimic** human reasoning and problem-solving skills by utilizing a vast **knowledge base** and **a set of rules or algorithms**."

Expert Systems



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MYCIN:

- One of the earliest **expert systems** in artificial intelligence,
- Developed in the mid-1970s at Stanford.
- It was designed to help doctors diagnose and recommend treatments for bacterial infections, especially blood infections and meningitis.
- MYCIN used **backward chaining** (goal-driven reasoning).
- One of MYCIN's innovations was its ability to **explain its reasoning.**

Knowledge Representation - MYCIN

- Storing medical knowledge as **production rules** (IF–THEN rules)
- Each rule had a **certainty factor (CF)**, representing how confident the system was in the conclusion.
- Example:

- *IF* the infection is primary-bacteremia *AND* the site of the culture is sterile *THEN*

there is suggestive evidence (0.7) that the infection is caused by enterobacteriaceae.

Example: MYCIN

A doctor is consulting MYCIN about a patient with a

MYCIN: What is the patient's age?

Doctor: 55

MYCIN: Has the patient had chills or shaking?

Doctor: Yes

MYCIN: What is the site of the culture?

Doctor: Blood

MYCIN: Was the stain gram-positive or gram-negative?

Doctor: Gram-negative

MYCIN: What is the morphology of the organism?

Doctor: Rod

Reasoning (internally using rules + certainty factors):
Gram-negative rod in blood → suggests enterobacteriaceae (CF 0.7)
Patient has chills → supports systemic infection
Age + site help refine probabilities

MYCIN (output):
The most likely causative organism is **E. coli** (certainty 0.7).
Other possible organisms include Klebsiella (certainty 0.4).
MYCIN (treatment recommendation):
Recommended therapy is **gentamicin** combined with **ampicillin**.
Dosage should be adjusted for body weight and renal function.

Example: MYCIN

- **Doctor:** Why do you recommend gentamicin?
- **MYCIN (explanation):**
- Because in cases of gram-negative rod septicemia, gentamicin is effective against enterobacteriaceae.
  Rule 215 applied with a certainty factor of 0.7.

Advantages of Expert Systems

Consistency

Always give the same advice for the same inputs.

Fast Decision Making

Processes information quickly using stored rules.

Knowledge Preservation

- Captures expertise of specialists and stores it permanently.
- Useful when human experts retire or are unavailable.

Explains reasoning

“why this conclusion was reached”

Accessibility

Provides expertise in remote areas where specialists are scarce

Disadvantages of Expert Systems

Limited Knowledge Base

f a new or rare situation arises outside the rule base, they may fail.

No Learning

Traditional expert systems don’t learn from experience unless updated manually.

Knowledge Acquisition Bottleneck

- Extracting and encoding expert knowledge into rules is difficult.
- Experts may disagree, and rules can become complex.

High Development Cost

Building and maintaining the knowledge base requires a lot of time and expertise.

No Common Sense

Cannot think outside programmed logic.
Struggle with unexpected scenarios or incomplete data.